

Primes Ending in 7

A special case of Dirichlet's Theorem

Gaia Corleto, Natalie Sin, Robert Rejna, Vishal Arun

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There are infinitely many primes $\equiv 1 \pmod{4}$ or $3 \pmod{4}$

- Define $A(x)$ and $B(x)$ such that:

$$A(x) = 1 \text{ for all } x.$$

$$B(x) = \begin{cases} 1 & \text{if } x \equiv 1 \pmod{4} \\ -1 & \text{if } x \equiv 3 \pmod{4} \end{cases}$$

- Construct indicator functions $f(x)$ and $g(x)$:

$$f(x) = \frac{A(x)+B(x)}{2} = \begin{cases} 1 & \text{if } x \equiv 1 \pmod{4} \\ 0 & \text{if } x \equiv 3 \pmod{4} \end{cases}$$

$$g(x) = \frac{A(x)-B(x)}{2} = \begin{cases} 0 & \text{if } x \equiv 1 \pmod{4} \\ 1 & \text{if } x \equiv 3 \pmod{4} \end{cases}$$

There are Infinitely Many Primes $\equiv 1 \pmod{4}$ or $3 \pmod{4}$

- Evaluate Dirichlet L-Series for $s > 1$:

$$L(s, A) = \sum_{n \text{ odd}} \frac{A(n)}{n^s} = \sum_{n \text{ odd}} \frac{1}{n^s}$$

$$L(s, B) = \sum_{n \text{ odd}} \frac{B(n)}{n^s} = 1 - \frac{1}{3^s} + \frac{1}{5^s} - \frac{1}{7^s} + \dots$$

- From Euler product formulas, we have:

$$L(s, \chi) = \prod_{p>2} \left(1 - \frac{\chi(p)}{p^s}\right)^{-1} \quad (\chi = A \text{ or } B)$$

- Taking the logarithm of both sides:

$$\log L(s, \chi) = \sum_{p>2} -\log \left(1 - \frac{\chi(p)}{p^s}\right)$$

There are infinitely many primes $\equiv 1 \pmod{4}$ or $3 \pmod{4}$

- Using the Taylor series expansion $-\log(1-x) = x + \frac{x^2}{2} + \frac{x^3}{3} + \dots$ where $x = \frac{\chi(p)}{p^s}$:

$$\begin{aligned}\log L(s, \chi) &= \sum_{p>2} \left(\frac{\chi(p)}{p^s} + \frac{\chi(p)^2}{2p^{2s}} + \frac{\chi(p)^3}{3p^{3s}} + \dots \right) \\ &= \sum_{p>2} \frac{\chi(p)}{p^s} + \sum_{p>2} \sum_{k=2}^{\infty} \frac{\chi(p)^k}{k \cdot p^{ks}}\end{aligned}$$

- Hence, $\log L(s, \chi) = \sum_{p>2} \frac{\chi(p)}{p^s} + O(1)$ as $s \rightarrow 1^+$, which gives:

$$\log L(s, A) \approx \sum_{p>2} \frac{A(p)}{p^s} = \sum_{p>2} \frac{1}{p^s}$$

$$\log L(s, B) \approx \sum_{p>2} \frac{B(p)}{p^s}$$

There are infinitely many primes $\equiv 1 \pmod{4}$ or $3 \pmod{4}$

$$L(s, A) = \sum_{n \text{ odd}} \frac{A(n)}{n^s} = \sum_{n \text{ odd}} \frac{1}{n^s} \quad L(s, B) = \sum_{n \text{ odd}} \frac{B(n)}{n^s} = 1 - \frac{1}{3^s} + \frac{1}{5^s} - \frac{1}{7^s} + \dots$$

Behavior of the series as $s \rightarrow 1^+$: $L(s, A)$: $L(s, A)$ behaves like the harmonic series

$$\implies \lim_{s \rightarrow 1^+} \log L(s, A) = \infty \quad \text{ $L(s, B)$: } L(1, B) = 1 - \frac{1}{3} + \frac{1}{5} - \frac{1}{7} + \dots = \frac{\pi}{4} \text{ (Leibniz's}$$

formula) Alternating series test implies convergence and $\pi/4 \neq 0$ implies $\log L(1, B)$ is a constant.

There are infinitely many primes $\equiv 1 \pmod{4}$ or $3 \pmod{4}$

- Combining via $f(x)$, $g(x)$:

For primes $\equiv 1 \pmod{4}$:

$$\sum_{p \equiv 1 \pmod{4}} \frac{1}{p^s} = \sum_{p > 2} \frac{f(p)}{p^s} = \frac{1}{2} \sum_{p > 2} \frac{A(p) + B(p)}{p^s}$$
$$\approx \frac{1}{2} \log L(s, A) + \frac{1}{2} \log L(s, B) \rightarrow \infty \quad \text{as } s \rightarrow 1^+.$$

For primes $\equiv 3 \pmod{4}$:

$$\sum_{p \equiv 3 \pmod{4}} \frac{1}{p^s} = \sum_{p > 2} \frac{g(p)}{p^s} = \frac{1}{2} \sum_{p > 2} \frac{A(p) - B(p)}{p^s}$$
$$\approx \frac{1}{2} \log L(s, A) - \frac{1}{2} \log L(s, B) \rightarrow \infty \quad \text{as } s \rightarrow 1^+.$$

7 mod 10

We can now consider the slightly more difficult special case of Dirichlet's theorem on primes in arithmetic sequences.

Theorem

There are infinitely many primes that end in a 7. Equivalently, there are infinitely many primes congruent to 7 mod 10.

The proof will involve several steps:

- Introducing Dirichlet L-series
- Examining behavior of these series as $s \rightarrow 1^+$
- Logarithms of the series
- A special linear combination

7 mod 10: Defining Dirichlet L-series

To simplify notation, we can choose primitive roots:

- 7 is a primitive root mod 10, namely $(\mathbb{Z}/10)^\times = \{1, 3, 7, 9\} = \{7^0, 7^3, 7^1, 7^2\}$.
- i is a primitive 4th root of unity, namely $\{1, i, -1, -i\} = \{i^0, i^1, i^2, i^3\}$.
(Note that $\phi(10) = 4$)

Now, as long as $n \bmod 10$ is a unit (i.e. $n \equiv 1, 3, 7, 9 \bmod 10$), we can define γ_n such that $n \equiv 7^{\gamma_n} \bmod 10$. (i.e. $\gamma_1 = 0, \gamma_3 = 3, \gamma_7 = 1, \gamma_9 = 2$)

We can also let i^k be an arbitrary 4th root of unity. ($k \in \{0, 1, 2, 3\}$)

(To shorten notation further, any unspecified congruences are mod 10)

7 mod 10: Defining Dirichlet L-series

Let $p \neq 2, 5$ and $s > 1$. Then by geometric series

$$\frac{1}{1 - i^{k\gamma_p} \frac{1}{p^s}} = 1 + i^{k\gamma_p} \frac{1}{p^s} + i^{2k\gamma_p} \frac{1}{p^{2s}} + i^{3k\gamma_p} \frac{1}{p^{3s}} + \dots$$

Now, take the product of these series over all primes $p \neq 2, 5$ for $k \in \{0, 1, 2, 3\}$.

$$\prod_{p \neq 2, 5} \frac{1}{1 - i^{k\gamma_p} \frac{1}{p^s}} = \sum_n i^{k\gamma_n} \frac{1}{n^s} = L_k \quad (1)$$

Where the sum is taken over $n \equiv 1, 3, 7, 9 \pmod{10}$. These are our Dirichlet L series.

7 mod 10: Defining Dirichlet L-series

To see the equality of product and sum, consider expanding the product on the RHS.

$$\prod_{p \neq 2,5} \frac{1}{1 - i^{k\gamma_p} \frac{1}{p^s}} = \prod_{p \neq 2,5} (1 + i^{k\gamma_p} \frac{1}{p^s} + i^{2k\gamma_p} \frac{1}{p^{2s}} + i^{3k\gamma_p} \frac{1}{p^{3s}} + \dots)$$

An arbitrary term of the expansion can be shown to equal $i^{k\gamma_n} \frac{1}{n^s}$ as in equation (1):

- Let $n = q_1^{m_1} q_2^{m_2} q_3^{m_3} \dots$, where $q_i \neq 2, 5$ are distinct primes and $m_i > 0$
- The n th term of the product is of the form $i^{k(m_1\gamma_{q_1} + m_2\gamma_{q_2} + m_3\gamma_{q_3} + \dots)} \frac{1}{n^s}$
- $m_1\gamma_{q_1} + m_2\gamma_{q_2} + m_3\gamma_{q_3} + \dots \equiv \gamma_n \pmod{4}$ by definition of γ_n
- Thus by Euler's theorem the n th term equals $i^{k\gamma_n} \frac{1}{n^s}$ as required

7 mod 10: The series L_0

Lemma

$L_0 \rightarrow \infty$ as $s \rightarrow 1^+$

Proof.

By (1) we have that

$$L_0 = \prod_{p \neq 2,5} \frac{1}{1 - \frac{1}{p^s}} = (1 - \frac{1}{2^s})(1 - \frac{1}{5^s}) \prod_p \frac{1}{1 - \frac{1}{p^s}} = (1 - \frac{1}{2^s})(1 - \frac{1}{5^s}) \sum_n \frac{1}{n^s}$$

Where the sum is now taken over all positive integers n .

As $s \rightarrow 1^+$, the RHS approaches the harmonic series, multiplied by a constant, hence it diverges to ∞ as required. □

7 mod 10: The series L_2

Lemma

L_2 converges to a positive real number as $s \rightarrow 1^+$.

Proof.

Consider when $s = 1$. Substituting γ_n in each case,

$$\begin{aligned} L_2 &= \sum_{n \equiv 1} \frac{1}{n} - \sum_{n \equiv 3} \frac{1}{n} - \sum_{n \equiv 7} \frac{1}{n} + \sum_{n \equiv 9} \frac{1}{n} \\ &= 1 - \frac{1}{3} - \frac{1}{7} + \frac{1}{9} + \frac{1}{11} - \frac{1}{13} - \frac{1}{17} + \frac{1}{19} + \dots \\ &= 1 - \frac{10}{21} + \frac{20}{99} - \frac{30}{221} + \dots \end{aligned}$$

Which converges by the alternating series test, hence L_0 converges when $s = 1$. The positiveness of the limit follows from the partial sums being always positive. □

7 mod 10: The series L_1 and L_3

Lemma

L_1 and L_3 converges to non-zero numbers as $s \rightarrow 1^+$.

Proof.

Again, consider when $s = 1$.

$$\begin{aligned} L_1 &= \sum_{n \equiv 1} \frac{1}{n} + i \sum_{n \equiv 3} \frac{1}{n} - i \sum_{n \equiv 7} \frac{1}{n} - \sum_{n \equiv 9} \frac{1}{n} = \sum_{n \equiv 1} \frac{1}{n} - \sum_{n \equiv 9} \frac{1}{n} + i \left(\sum_{n \equiv 3} \frac{1}{n} - \sum_{n \equiv 7} \frac{1}{n} \right) \\ L_3 &= \sum_{n \equiv 1} \frac{1}{n} - i \sum_{n \equiv 3} \frac{1}{n} + i \sum_{n \equiv 7} \frac{1}{n} - \sum_{n \equiv 9} \frac{1}{n} = \sum_{n \equiv 1} \frac{1}{n} - \sum_{n \equiv 9} \frac{1}{n} - i \left(\sum_{n \equiv 3} \frac{1}{n} - \sum_{n \equiv 7} \frac{1}{n} \right) \end{aligned}$$

Hence $L_3 = \overline{L_1}$, so it is enough to consider the convergence of L_1 .

7 mod 10: The series L_1 and L_3

Proof (Continued).

Consider the real and imaginary parts of L_1 separately as follows

$$\operatorname{Re}(L_1) = \sum_{n \equiv 1} \frac{1}{n} - \sum_{n \equiv 9} \frac{1}{n} = 1 - \frac{1}{9} + \frac{1}{11} - \frac{1}{9} + \frac{1}{21} - \frac{1}{29} + \dots$$

$$\operatorname{Im}(L_1) = \sum_{n \equiv 3} \frac{1}{n} - \sum_{n \equiv 7} \frac{1}{n} = \frac{1}{3} - \frac{1}{7} + \frac{1}{13} - \frac{1}{17} + \frac{1}{23} - \frac{1}{27} + \dots$$

These converge by the alternating series test, hence L_1 and L_3 converge when $s = 1$. It remains to show that the limits of L_1 and L_3 are non-zero. Consider the product

$$L_1 L_3 = L_1 \overline{L_1} = \operatorname{Re}(L_1)^2 + \operatorname{Im}(L_1)^2 \geq 0$$

To show strictness, consider that the partial sums of $\operatorname{Re}(L_1)$ are strictly positive, hence $\operatorname{Re}(L_1) > 0$ and thus $L_1 L_3 > 0$. In particular, this means that L_1 and L_3 are non-zero. $\square$

7 mod 10: Logarithms of L_k

For $s > 1$

$$\log L_k = \log \prod_{p \neq 2,5} \frac{1}{1 - i^{k\gamma_p} \frac{1}{p^s}} = \sum_{p \neq 2,5} \log \frac{1}{1 - i^{k\gamma_p} \frac{1}{p^s}}$$

We can then apply Taylor's theorem and rearrange to get

$$\begin{aligned} \log L_k &= \sum_{p \neq 2,5} \left(i^{k\gamma_p} \frac{1}{p^s} + \frac{1}{2} i^{2k\gamma_p} \frac{1}{p^{2s}} + \frac{1}{3} i^{3k\gamma_p} \frac{1}{p^{3s}} + \dots \right) \\ &= \sum_{p \neq 2,5} i^{k\gamma_p} \frac{1}{p^s} + \frac{1}{2} \sum_{p \neq 2,5} i^{2k\gamma_p} \frac{1}{p^{2s}} + \frac{1}{3} \sum_{p \neq 2,5} i^{3k\gamma_p} \frac{1}{p^{3s}} + \dots \end{aligned} \tag{2}$$

This rearranging is justified since for $s > 1$, convergence of L_k is guaranteed. Also, this is well defined (taking the principal branch) since as $s \rightarrow 1^+$, L_0 and L_2 are real and non-zero and L_1 and L_3 have positive real parts.

7 mod 10: The proof

Applying our result from (2) to the linear combination below, we get

$$\begin{aligned}\log L_0 + i^{-1} \log L_1 + i^{-2} \log L_2 + i^{-3} \log L_3 &= \sum_{p \neq 2,5} (1 + i^{\gamma_p-1} + i^{2(\gamma_p-1)} + i^{3(\gamma_p-1)}) \frac{1}{p^s} \\ &\quad + \frac{1}{2} \sum_{p \neq 2,5} (1 + i^{2\gamma_p-1} + i^{2(2\gamma_p-1)} + i^{3(2\gamma_p-1)}) \frac{1}{p^{2s}} \\ &\quad + \frac{1}{3} \sum_{p \neq 2,5} (1 + i^{3\gamma_p-1} + i^{2(3\gamma_p-1)} + i^{3(3\gamma_p-1)}) \frac{1}{p^{3s}} \\ &\quad + \dots\end{aligned}\tag{3}$$

Note that $1 + i^{h\gamma_p-1} + i^{2(h\gamma_p-1)} + i^{3(h\gamma_p-1)} = 0$ except when $h\gamma_p - 1 \equiv 0 \pmod{4}$ where it is 4. Also, $h\gamma_p - 1 \equiv 0 \pmod{4} \iff p^h \equiv 7 \pmod{10}$.

7 mod 10: The proof

Thus, we can simplify the RHS of equation (3) to get

$$\frac{1}{4}(\log L_0 + i^{-1} \log L_1 + i^{-2} \log L_2 + i^{-3} \log L_3) = \sum_{p \equiv 7} \frac{1}{p^s} + \frac{1}{2} \sum_{p^2 \equiv 7} \frac{1}{p^{2s}} + \frac{1}{3} \sum_{p^3 \equiv 7} \frac{1}{p^{3s}} + \dots \quad (4)$$

Now, we compare sides of equation (4) as $s \rightarrow 1^+$. From our lemmas

- $\log L_0 \rightarrow \infty$ as $s \rightarrow 1^+$
- $\log L_1$, $\log L_2$, and $\log L_3$ remain finite
- Also, all terms on the RHS except the leading term remain finite
(compare with $\sum_n \frac{1}{n^m}$, $m = 2, 3, \dots$)

Hence, we can conclude that the term $\sum_{p \equiv 7} \frac{1}{p^s} \rightarrow \infty$ as $s \rightarrow 1^+$. So, there must be infinitely many primes congruent to 7 mod 10.

Generalisation to Dirichlet's Theorem

Theorem (Dirichlet's Theorem)

Let a and d be positive coprime integers. Then, there exist infinitely many prime numbers p such that:

$$p \equiv a \pmod{d}$$

- For the 7 mod 10 group, we used the integers coprime to 10 which were 1,3,7,9 which form a multiplicative group of 10. To find how many numbers less than d , coprime to d we can use the Euler Totient Function $\phi(d)$
- Instead of using the 4th roots of unity we now need $\phi(d)$ characters, so we can generalise by using the $\phi(d)^{th}$ roots of unity.
- We now have our **L-series**: $L(s, \chi) = \sum_{n=1}^{\infty} \frac{\chi(n)}{n^s}$. We need a way to make a linear combination of them such that all the complex numbers cancel out, except for $p \equiv a \pmod{d}$, like we did to have just $p \equiv 7 \pmod{10}$ left.

Dirichlet Characters

To generalize our filtering mechanism, we must formally define a **Dirichlet character** (χ).

- A character is a **group homomorphism** from the coprime residue classes modulo d to the complex numbers. It preserves multiplication:

$$\chi(a \cdot b) = \chi(a) \cdot \chi(b)$$

- Because of this structure, characters must map the elements to the complex $\phi(d)^{\text{th}}$ roots of unity (along with mapping non-coprime elements to 0).
- In our 7 (mod 10) case, our characters mapped the group $\{1, 3, 7, 9\}$ strictly to the 4th roots of unity:

$$\{1, -1, i, -i\}$$

- This structure provides the precise periodic cancellations needed to construct our prime-isolating filter.

Filtering Out Primes: $p \not\equiv a \pmod{d}$

To isolate our target residue class a and eliminate all other columns, we analyze the following expression:

$$\overline{\chi}(a) \cdot \chi(p) = \frac{\chi(p)}{\chi(a)} = \chi(p \cdot a^{-1})$$

- If p is in the target residue class a , then $p \equiv a$, so the expression simplifies cleanly to $\chi(a \cdot a^{-1}) = \chi(1) = 1$. Since every character maps $1 \rightarrow 1$ by definition, summing across all characters yields a total value of $\phi(d)$.
- If p is not in the target residue class a , then the expression reduces to $\chi(g)$ where $g \not\equiv 1 \pmod{d}$. The linear combination cycles through the roots of unity, balancing symmetrically and causing the entire sum to collapse to 0.

Building the Proof

- Mirroring our $7 \pmod{10}$ approach, we take the logarithms of the general L -series products and divide by $\phi(d)$ to construct our linear combination:

$$\frac{1}{\phi(d)} \sum_{\chi} \bar{\chi}(a) \log L(s, \chi) \approx \sum_{\substack{p \equiv a \\ (\text{mod } d)}} \frac{1}{p^s}$$

- This gives us our final form. Because the principal character series blows up to infinity while all non-principal components remain finite, this forcing function proves that the sequence contains an infinite number of primes.
- For brevity, this structural sketch assumes the non-vanishing property $L(1, \chi) \neq 0$ for all non-principal characters, ensuring that taking the logarithm at the boundary edge is mathematically well-defined.

Dirichlet Density

- If we were to pick a prime number at random, how likely is it to end in 7?
- As we are dealing with sets of infinite size, we need a new way to measure this.

Definition

Let A be a set of prime numbers. The **Dirichlet density** of A , denoted by $d(A)$, is defined as the limit:

$$d(A) = \lim_{s \rightarrow 1^+} \frac{\sum_{p \in A} \frac{1}{p^s}}{\sum_p \frac{1}{p^s}} \quad (5)$$

provided that the limit exists.

Here, the numerator acts as our target filter, summing $1/p^s$ exclusively over the primes belonging to the chosen subset A .

We will now give a proof of the distribution of the primes ending in 7.

Proof of Dirichlet Density mod 10

Returning to our previous calculations:

$$\log L_k(s) = - \sum_{p \neq 2,5} \log \left(1 - \frac{c_k(p)}{p^s} \right) = \sum_{p \neq 2,5} \frac{c_k(p)}{p^s} + \mathcal{O}(1) \quad (6)$$

where $c_k(p)$ represents the specific sequence coefficients $(\pm 1, \pm i)$ assigned to each residue class.

- L_0 : Recalling previous steps and using $\zeta(s) = \sum_{n=1}^{\infty} \frac{1}{n^s}$:

$$\log L_0(s) = \log \zeta(s) + \log(1 - 2^{-s}) + \log(1 - 5^{-s}) = \log \zeta(s) + \mathcal{O}(1)$$

As $s \rightarrow 1^+$, $\zeta(s)$ tends to the harmonic series, hence $\log L_0(s) \rightarrow \infty$.

- L_1, L_2, L_3 : We earlier established the convergence of these series:

$$\lim_{s \rightarrow 1^+} \log L_k(s) = C_k \quad (\text{where } C_k \in \mathbb{C} \text{ is a finite constant})$$

Proof Continued

- We define 4 sums based on all possible residue classes mod 10 (excluding $p = 2, 5$)

$$A(s) = \sum_{p \equiv 1} \frac{1}{p^s}, \quad B(s) = \sum_{p \equiv 3} \frac{1}{p^s}, \quad C(s) = \sum_{p \equiv 7} \frac{1}{p^s}, \quad D(s) = \sum_{p \equiv 9} \frac{1}{p^s}$$

- We expand $\log L_k$ explicitly using the sequence coefficients ($c_k = \pm 1, \pm i$) to obtain the system of equations:

$$L_0 : A(s) + B(s) + C(s) + D(s) = \ln \zeta(s) + \mathcal{O}(1)$$

$$L_1 : A(s) - B(s) - C(s) + D(s) = C_2$$

$$L_2 : A(s) + iB(s) - iC(s) - D(s) = C_3$$

$$L_3 : A(s) - iB(s) + iC(s) - D(s) = C_4$$

- We then solve for $C(s)$ using the combination: $\mathbf{1} \cdot \mathbf{L}_0 - \mathbf{1} \cdot \mathbf{L}_1 + \mathbf{i} \cdot \mathbf{L}_2 - \mathbf{i} \cdot \mathbf{L}_3$.
- The coefficients for A, B , and D perfectly vanish to 0, leaving:

$$4C(s) = \ln \zeta(s) + \mathcal{O}(1) \implies C(s) = \frac{1}{4} \ln \zeta(s) + \mathcal{O}(1)$$

Conclusion on the Distribution of Primes ending in 7

- By definition, the density $d(P_7)$ measures our target prime reciprocal sum against the global baseline of all primes as $s \rightarrow 1^+$:

$$d(P_7) = \lim_{s \rightarrow 1^+} \frac{C(s)}{\sum_p \frac{1}{p^s}} = \lim_{s \rightarrow 1^+} \frac{\frac{1}{4} \ln \zeta(s) + \mathcal{O}(1)}{\ln \zeta(s) + \mathcal{O}(1)}$$

- Dividing both the numerator and denominator by $\ln \zeta(s)$, the bounded error remainders vanish to 0 since $\ln \zeta(s) \rightarrow \infty$ as $s \rightarrow 1^+$:

$$d(P_7) = \lim_{s \rightarrow 1^+} \frac{\frac{1}{4} + \frac{\mathcal{O}(1)}{\ln \zeta(s)}}{1 + \frac{\mathcal{O}(1)}{\ln \zeta(s)}} = \frac{\frac{1}{4} + 0}{1 + 0} = \boxed{\frac{1}{4}} = \frac{1}{\phi(10)}$$

Conclusion: Similar calculations for the other 3 sums would show that primes are perfectly **equidistributed** across all four allowed residue classes modulo 10:

$$d(P_1) = 25\%, \quad d(P_3) = 25\%, \quad \mathbf{d(P_7) = 25\%}, \quad d(P_9) = 25\%$$